

Sub-Lemma Z reduces to (L1)–(L4): the written reduction

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Scripts: proofs/scripts/day205/reduction*.py (run reduction_run.all.sh; logs alongside)

0. Verdict

The reduction holds, with no gap. Every step is an elementary identity with a short proof, and each one is also checked by an exact script. Together with your Theorem 3 ((L1)–(L4) for all $m \geq 1$, $r \geq 1$), it proves Sub-Lemma Z as a polynomial identity in Λ_m for *every* $m \geq 1$ and $r \geq 1$. The “exactly four nonzero e -coefficients” form holds for $r \geq 2$, $m \geq r + 2$ (and in the stable limit $\Lambda_{q,t}$). The reduction depends on one outside input: the definition of Hikita’s $\star$ (Def. 3.4) and the bijectivity of $\mathfrak{q}_{(m)}$. I cite these and do not re-prove them. The reduction does *not* use your Λ_m -linearity (Lemma 5), $\star$ -associativity, Theorem 3.12, or $m \geq r + 2$.

The Day 204 sketch had three imprecisions, all harmless and corrected here (§5).

1 Definitions (matching the code)

Fix $m \geq 1$. Work over $\mathbb{Q}(q, t)$ with $x = (X_1, \dots, X_m)$. Let $\Lambda_m = \mathbb{Q}(q, t)[x]^{S_m}$, $e_k = e_k(X_1, \dots, X_m)$ ($e_0 = 1$, $e_k = 0$ for $k < 0$ or $k > m$), $e_\mu := \prod_i e_{\mu_i}$ (ordinary product; in particular $e_{(r,1)} = e_r e_1$). The *tail* is $\hat{x}_1 = (X_2, \dots, X_m)$, and $e_k(\hat{x}_1)$ uses the same conventions with $m - 1$ variables. $[n]_t = 1 + t + \dots + t^{n-1}$ for $n \geq 1$, and $[n]_t = 0$ for $n \leq 0$.

Definition 1 (level-one polynomial representation; build_action in scripts/day198/p2Y_er.py).

For $1 \leq i \leq m - 1$, with s_i the transposition $X_i \leftrightarrow X_{i+1}$,

$$T_i F = t s_i F + (t - 1) X_{i+1} \frac{s_i F - F}{X_i - X_{i+1}} = t F + \frac{t X_i - X_{i+1}}{X_i - X_{i+1}} (s_i F - F), \quad (1)$$

$$T_i^{-1} = t^{-1} T_i - (1 - t^{-1}), \quad \pi F = X_1 F(X_2, \dots, X_m, q^{-1} X_1),$$

$$Y_i = t^{m-i} T_{i-1} \cdots T_1 \pi T_{m-1}^{-1} \cdots T_i^{-1} \quad (\text{rightmost acts first}), \quad e_1(Y) = \sum_{i=1}^m Y_i.$$

Write $G \bullet F$ for this action. The two forms in (1) are equal: the first is the code, the second is yours (check: reduction_0 (b)). The T_i satisfy $(T_i - t)(T_i + 1) = 0$ and the braid relations (reduction_0 (c), random inputs). All operators are $\mathbb{Q}(q, t)$ -linear.

Definition 2 (partial symmetrizer). $\rho_j := T_{j-1} T_{j-2} \cdots T_1$ ($\rho_1 = \text{id}$), and $\sigma_m := \sum_{k=0}^{m-1} T_k T_{k-1} \cdots T_1 = \sum_{j=1}^m \rho_j$.

Definition 3 (Hikita's $\star$, arXiv:2503.23597 Def. 3.4). $\mathfrak{q}_{(m)}(F(Y)) := F(Y) \bullet 1$, a linear isomorphism from symmetric polynomials in Y onto Λ_m (Hikita; *cited, not re-proved*). For $F, G \in \Lambda_m$, $F \star G := \mathfrak{q}_{(m)}(\mathfrak{q}_{(m)}^{-1}(F) \mathfrak{q}_{(m)}^{-1}(G))$.

Sub-Lemma Z (Day 203). $Z_r := e_1 \star e_{(r,1)}$. The claim is

$$Z_r = \frac{(q-1)^2[r+2]_t}{q^2} e_{r+2} + \frac{(q-1)(qt[r]_t + 1)}{q^2} e_{(r+1,1)} + \frac{(q-1)[2]_t}{q^2} e_{(r,2)} + \frac{1}{q^2} e_{(r,1,1)}. \quad (\text{Z})$$

Imported hypotheses (your Theorem 3). For all $m \geq 1$ and $r \geq 1$, in Λ_m :

- (L1) $\sigma_m[X_1 e_r(\hat{x}_1) e_1(\hat{x}_1)] = [r+2]_t e_{r+2} + t[r]_t e_{(r+1,1)},$
- (L2) $\sigma_m[X_1^2 e_r(\hat{x}_1)] = -[r+2]_t e_{r+2} + e_{(r+1,1)},$
- (L3) $\sigma_m[X_1^2 e_{r-1}(\hat{x}_1) e_1(\hat{x}_1)] = -[r+2]_t e_{r+2} - t[r]_t e_{(r+1,1)} + [2]_t e_{(r,2)},$
- (L4) $\sigma_m[X_1^3 e_{r-1}(\hat{x}_1)] = [r+2]_t e_{r+2} - e_{(r+1,1)} - [2]_t e_{(r,2)} + e_{(r,1,1)}.$

2 The reduction

Lemma 4 (Step R0: $\star$ to $e_1(Y)$). $Y_i \bullet 1 = X_i$ for all i . Hence $e_1(Y) \bullet 1 = e_1$, and for every $G \in \Lambda_m$,

$$e_1 \star G = e_1(Y) \bullet G. \quad \text{In particular} \quad Z_r = e_1(Y) \bullet (e_r e_1).$$

Proof. From (1) with $F = X_j$: $s_j F = X_{j+1}$, so $T_j X_j = t X_{j+1} + (t-1) X_{j+1} \frac{X_{j+1} - X_j}{X_j - X_{j+1}} = X_{j+1}$. By Lemma 5 below, applied to $F = 1$, we get $Y_i \bullet 1 = \rho_i \pi(1) = \rho_i X_1 = X_i$. Summing gives $e_1(Y) \bullet 1 = e_1$, so $\mathfrak{q}_{(m)}^{-1}(e_1) = e_1(Y)$. Let $B := \mathfrak{q}_{(m)}^{-1}(G)$, so $B \bullet 1 = G$. Since $\bullet$ is a module action, $e_1 \star G = \mathfrak{q}_{(m)}(e_1(Y)B) = (e_1(Y)B) \bullet 1 = e_1(Y) \bullet (B \bullet 1) = e_1(Y) \bullet G$. $\square$

Only the case $a = 1$ of the Day 200 intertwiner is needed, and it is proved above. The general $t^{(a)}$ factor is not used.

Lemma 5 (Step A: $e_1(Y) = \sigma_m \pi$ on Λ_m). For $F \in \Lambda_m$ and $1 \leq i \leq m$: $Y_i \bullet F = \rho_i \pi F$. Hence

$$e_1(Y) \bullet F = \sigma_m \pi F \quad (F \in \Lambda_m, \text{ any degree, any } m).$$

Proof. If $s_j F = F$, then (1) gives $T_j F = tF$, and so $T_j^{-1} F = t^{-1} \cdot tF - (1 - t^{-1})F = t^{-1} F$. The result $t^{-1} F$ is still symmetric, so by induction $T_{m-1}^{-1} \cdots T_i^{-1} F = t^{-(m-i)} F$. Therefore $Y_i \bullet F = t^{m-i} \rho_i \pi(t^{-(m-i)} F) = \rho_i \pi F$ by linearity. Sum over i . $\square$

Symmetry of F is essential. For $F = X_1^2 X_3$ ($m = 3$) the per- i identity fails (**reduction A**, negative control).

Lemma 6 (Step B: the π -split). Let ρ_q be the $\mathbb{Q}(q, t)$ -algebra endomorphism $X_j \mapsto X_{j+1}$ ($j < m$), $X_m \mapsto q^{-1} X_1$. Then $\pi F = X_1 \rho_q(F)$, and so

$$X_1 \pi(FG) = \pi(F) \pi(G) \quad \text{for all } F, G. \quad (2)$$

With $f := e_r(\hat{x}_1)$, $g := e_{r-1}(\hat{x}_1)$, $h := e_1(\hat{x}_1)$ and $r \geq 1$:

$$\pi(e_r) = X_1 f + q^{-1} X_1^2 g, \quad \pi(e_r e_1) = X_1 f h + q^{-1} X_1^2 (f + g h) + q^{-2} X_1^3 g. \quad (3)$$

Proof. $\pi = X_1 \rho_q$ is the definition. Since ρ_q is multiplicative, $X_1 \pi(FG) = X_1^2 \rho_q(F) \rho_q(G) = \pi(F) \pi(G)$. Next, $\rho_q(e_r) = e_r(X_2, \dots, X_m, q^{-1} X_1) = e_r(\hat{x}_1) + q^{-1} X_1 e_{r-1}(\hat{x}_1)$, using $e_r(\text{set} \cup \{z\}) = e_r(\text{set}) + z e_{r-1}(\text{set})$. So $\rho_q(e_r e_1) = (f + q^{-1} X_1 g)(h + q^{-1} X_1)$. Multiply out and then multiply by X_1 . $\square$

The Day 204 phrase “ $\pi(FG) = \pi(F)\pi(G)/X_1$ ” is (2), and it holds because ρ_q is a ring homomorphism.

Step C (linearity). Each T_i is $\mathbb{Q}(q, t)$ -linear, so σ_m is too, and the weights q^{-a} are scalars. Lemmas 4, 5 and 6 give

$$Z_r = \sigma_m[X_1 f h] + q^{-1} \sigma_m[X_1^2 f] + q^{-1} \sigma_m[X_1^2 g h] + q^{-2} \sigma_m[X_1^3 g]. \quad (4)$$

The four brackets are exactly the left-hand sides of (L1), (L2), (L3), (L4). The weights are $(1, q^{-1}, q^{-1}, q^{-2})$.

Step D (assembly, symbolic in r). Substitute (L1)–(L4) into (4) and collect terms. Write $A = [r+2]_t$, $B = [r]_t$, $D = [2]_t$. These can be treated as independent indeterminates, since each coefficient below is linear in them. The identity is therefore valid for every integer $r \geq 1$ without substituting $u = t^r$:

$$\begin{aligned} [e_{r+2}] : \quad & A(1 - q^{-1} - q^{-1} + q^{-2}) = (1 - q^{-1})^2 A = \frac{(q-1)^2}{q^2} [r+2]_t, \\ [e_{(r+1,1)}] : \quad & tB + q^{-1} - q^{-1} tB - q^{-2} = (1 - q^{-1})(tB + q^{-1}) = \frac{(q-1)(qt[r]_t + 1)}{q^2}, \\ [e_{(r,2)}] : \quad & q^{-1} D - q^{-2} D = \frac{(q-1)[2]_t}{q^2}, \\ [e_{(r,1,1)}] : \quad & q^{-2}. \end{aligned}$$

These are exactly the four coefficients of (Z).

Theorem 7. Assume (L1)–(L4) hold in Λ_m for a given $m \geq 1$ and $r \geq 1$. Then (Z) holds in Λ_m as a polynomial identity. For $r = 1$ the terms $e_{(r,2)}$ and $e_{(r+1,1)}$ are the same element $e_{(2,1)}$, and their coefficients add. Suppose in addition $r \geq 2$ and $m \geq r+2$. Then the partitions $(r+2), (r+1,1), (r,2), (r,1,1)$ are distinct, all have largest part $\leq m$, and $\{e_\mu : \mu \vdash r+2, \mu_1 \leq m\}$ is linearly independent in Λ_m (these are distinct monomials in $e_1, \dots, e_m$, which are algebraically independent: Macdonald I, (2.4)). So (Z) is the e -expansion of Z_r , and it has exactly four nonzero coefficients. By your Theorem 3, the hypothesis holds for all $m \geq 1, r \geq 1$. Because the identity holds at every m with m -independent coefficients, it also holds in the stable limit.

Proof. Combine Lemma 4, Lemma 5, (3), (4), (L1)–(L4), and Step D. $\square$

Remark 8 (same mechanism, one level down: Hikita’s Thm 3.12). By Lemmas 5 and 6, $e_1 \star e_r = \sigma_m[X_1 e_r(\hat{x}_1)] + q^{-1} \sigma_m[X_1^2 e_{r-1}(\hat{x}_1)]$. Your (5) and (6) give $[r+1]_t e_{r+1} + q^{-1}(-[r+1]_t e_{r+1} + e_1 e_r)$, which is Hikita’s Theorem 3.12. This ties the code’s conventions to a published theorem (`reduction_0` (e): exact match for $r = 1..7, m = 2..r+3$).

3 Verification (all exact; integer Laurent coefficients in t, q^{-1})

`reduction_engine.py` is a new, dependency-free implementation. It is cross-checked against the Day 198 SymPy `build_action`, which is the code that checked Sub-Lemma Z, for T_i, T_i^{-1}, π, Y_i

on random inputs with $m = 2, 3, 4$ (**reduction_0** (a)). All checks are polynomial identities in $X_1, \dots, X_m$. None of them relies on the uniqueness of an e -expansion, except where marked.

Script	What	Range
reduction_0_conventions	(a) engine = Day 198 code; (b) two T_i forms; (c) Hecke/braid; (d) $Y_i \bullet 1 = X_i$, $T_i F = tF$; (e) Thm 3.12 anchor	$m \leq 6$; (e) $r \leq 7$
reduction_A_sigma_pi	$Y_i F = \rho_i \pi F$ for each i ; $F = e_r e_1$, e_r ; neg. control	$r = 2..7$, $m = r+2..r+4$ (≤ 10)
reduction_B_pi_split	(2), (3)	$r = 1..7$, $m = 2..r+4$ (≤ 10)
reduction_C_pieces	(4) as polynomials; (L1)–(L4) as polynomials, q -free; unique e -expansion when $m \geq r + 2$	$r = 1..7$, $m = 2..r+4$ (≤ 10)
reduction_D_assembly	Step D symbolic (A, B, D free; also $u = t^r$); end-to-end $e_1(Y) \bullet (e_r e_1)$ vs (Z); support size; neg. controls	$r = 1..7$, $m = 2..r+4$ (≤ 10)

All five scripts report **PASSED**. The negative controls behave correctly: wrong weights $(1, q^{-1}, q^{-1}, q^{-1})$ are rejected, a perturbed (L2) is rejected, and a non-symmetric F breaks Step A. The (L1)–(L4) checks in **reduction_C** are a sanity check on the statement being imported. They are not part of the reduction.

4 Grades

Step	Content	Grade
R0	$e_1 \star G = e_1(Y) \bullet G$; $Y_i \bullet 1 = X_i$	proved , given Hikita Def. 3.4 and bijectivity of $q_{(m)}$ (cited)
A	$e_1(Y) \bullet F = \sigma_m \pi F$, $F \in \Lambda_m$	proved + computed
B	π -split (3)	proved + computed
C	four-piece decomposition (4)	proved (linearity) + computed
D	assembly, symbolic in r	proved (4 lines) + symbolic check
Uniqueness / support	$m \geq r + 2$, $r \geq 2$	proved (Macdonald I (2.4))
Reduction as a whole	(L1)–(L4) $\Rightarrow$ (Z)	proved

No step is only **computed**.

Weakest link. This is R0’s interface with Hikita, not the algebra of the reduction. There are two parts: (i) the bijectivity of $q_{(m)}$ is cited, not re-proved; (ii) the code’s Y_i and π are assumed to be Hikita’s eq. (3). Part (ii) is supported by **reduction_0** (e), which reproduces Thm 3.12 exactly, and by Day 190, which reproduces his Ex. 4.6. It is not a line-by-line comparison with the paper. If there were a convention mismatch, (Z) would still be proved for the operator $e_1(Y) \bullet$ that is actually implemented, and this is the object that every Sub-Lemma Z computation has used.

5 Corrections to the Day 204 sketch

1. Step A was stated “on level-one polynomials of degree $\leq m$ ” and was only checked numerically. The correct scope is all *symmetric* F , with no degree bound. It is false for non-symmetric F . It is now proved (Lemma 5).
2. The sketch said (L1)–(L4) “live at $m \geq r + 2$ ”. Neither the reduction nor your identities need this. The hypothesis $m \geq r + 2$ enters only when reading off a unique e -expansion.
3. $\pi(FG) = \pi(F)\pi(G)/X_1$ was “verified numerically”. It is now proved: ρ_q is a ring homomorphism.

Consequence. Sub-Lemma Z is **proved** for all $r \geq 1$ and $m \geq 1$ as an identity, with unique four-term support for $r \geq 2$, $m \geq r + 2$. This rests on (L1)–(L4) [Clio, Theorem 3] and on Hikita’s Def. 3.4 setup. Before any registry promotion, your check of this note is the remaining peer step.